

EXECUTIVE BUSINESS REPORT

End-to-End Sales Forecasting & Demand Intelligence System

Superstore Retail Sales, 2015–2018

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July 2026

1. Executive Summary

Superstore's sales history from January 2015 to December 2018, comprising 9,800 order line items across 4,922 orders and \$2.26 million in revenue, was analyzed to answer four key inventory planning questions: what demand is likely next quarter, which weeks represent genuine anomalies, how products should be stocked differently, and how much confidence decision-makers can place in those forecasts.

Revenue grew 50.5% from 2015 to 2018, driven overwhelmingly by a September, November, and December surge that repeats in every year of the data. Technology is the strongest category (36.6% of revenue) and West the strongest region, though East grows the most consistently of the four. A statistical forecasting model (SARIMA) was selected for production use after being tested against two alternatives on a rigorous backtest, and it projects total sales of roughly \$50,100 in January 2019, \$35,700 in February, and \$69,700 in March, each with a calibrated confidence range.

Two independent anomaly-detection methods flagged 19 unusual sales weeks out of 209 analyzed, with 3 weeks flagged by both methods — the highest-confidence signals in the data. Products were grouped into four demand segments, each with materially different stocking needs, from the eight-category revenue backbone that accounts for 73.2% of total sales to a single volatile outlier (Supplies) whose apparent growth turns out to be a one-time rebound rather than a trend.

The recommendations that follow are built entirely from this analysis: segment-specific inventory policy, a production forecasting workflow, a tiered anomaly-alert design, and an honest account of what this system cannot yet do. Full detail, including every chart and calculation behind these numbers, is available in the accompanying analysis notebook and interactive dashboard.

2. Business Problem

Every retail business runs into the same tension: order too much of a product and capital sits on a warehouse shelf; order too little and a sale walks out the door to a competitor. Getting this right depends on knowing, with some confidence, how demand for each product is likely to move — and demand rarely moves in a straight line. It swings with the seasons, differs by region, occasionally spikes or drops for reasons worth understanding, and behaves very differently for a \$15 box of paper than it does for a \$2,000 copier.

Superstore's leadership needs a way to see these patterns clearly: a reliable forecast of near-term sales, an early warning system for weeks that don't fit the expected pattern, and a way to group thousands of individual products into a manageable number of stocking strategies rather than making one-off judgment calls. This project builds that system directly from four years of the company's own order history.

3. Project Objectives

- Build a clean, quality-checked dataset from four years of raw order data, with every transformation verified rather than assumed.
- Characterize the underlying structure of sales over time — trend, seasonality, and whether the series behaves predictably enough to forecast.

- Build and fairly compare multiple forecasting approaches, and select one for production use based on evidence rather than preference.
- Extend that forecast to the individual product categories and regions that matter most to planning.
- Detect unusual sales weeks using two independent methods and reconcile where they agree and disagree.
- Segment products by demand behavior so that stocking policy can be tailored to how each group actually sells.
- Package the results in an interactive dashboard so the findings remain usable after this report is read once and filed away.

4. Dataset Overview

The analysis is built on the Superstore Sales dataset (sourced from Kaggle), covering 9,800 order line items across 4,922 unique orders, from January 3, 2015 to December 30, 2018 — four complete calendar years. Each line item records a product sale with its order and ship date, customer segment, region, product category and sub-category, and sale value.

The data spans 3 top-level product categories (Technology, Furniture, Office Supplies) broken into 17 sub-categories, and 4 US regions (Central, East, South, West).

A second, supplementary dataset (Video Game Sales) was evaluated for possible use in the anomaly-detection phase of this project, as suggested in the original project brief. It was deliberately not used: the two datasets share no common product, order, or geography identifier, so any merge would have been a forced concatenation rather than a meaningful analytical join. This is noted here for transparency, since the decision not to use available data was a considered one rather than an oversight.

5. Data Preparation

Order and ship dates in the raw file are stored day-first (DD/MM/YYYY), a format that pandas cannot always infer safely — several dates in the data, such as 08/11/2017, are valid under both day-first and month-first readings. The format was pinned explicitly during loading to avoid a silent misparse that would have corrupted every downstream time-based calculation.

A structured data quality audit followed: checking column types, missing values, duplicate records, and the internal consistency of the order-to-ship timeline. The dataset held up well. The only missing values were in the Postal Code field (11 of 9,800 rows, all from Vermont — a known quirk where Vermont ZIP codes with a leading zero are dropped on import), and that field plays no role in the sales analysis. There were no duplicate records and no order that shipped before it was placed.

Calendar features — year, month, ISO week number, day of week, quarter, and season — were engineered from the order date, along with a fulfillment-time measure (ship date minus order date). Seasons were assigned using the standard meteorological calendar, appropriate since all orders ship within the United States.

The line-item data was then aggregated into daily, weekly, and monthly totals. Each aggregation was checked to confirm it preserved the full \$2,261,537 in revenue with no loss — aggregation should never silently drop data, and this was verified rather than assumed.

6. Exploratory Data Analysis

Four questions were asked of the cleaned data, each answered with evidence rather than assumption.

Which category generates the most revenue?

Technology leads with \$827,456 (36.6% of total revenue), ahead of Furniture (\$728,659, 32.2%) and Office Supplies (\$705,422, 31.2%). The three categories are fairly balanced, but Technology's higher average order value gives it the edge despite not having the most individual sales.

Which region grows most consistently?

West generates the most total revenue (\$710,220 over four years), but East is the most consistent grower — the only region to increase every single year, tracking a nearly straight-line trend ($R^2 \approx 0.997$). West grows faster on average but unevenly, including an actual dip in 2016.

How long does fulfillment take, and does it vary by region?

Orders ship in 3.96 days on average, with almost no regional variation — from 3.91 days in the East to 4.07 days in Central, a spread well within normal day-to-day variation. Fulfillment speed is effectively a national constant, not a regional differentiator.

Is there a repeatable seasonal pattern?

Yes, and it is strong. September, November, and December rank among each year's three highest-selling months in every one of the four years studied, while January and February are consistently the weakest. This pattern — a September back-to-school bump followed by the November–December holiday peak — turns out to be the single most important signal in the entire dataset, more important than the underlying growth trend, as the next section shows.

7. Time Series Analysis

Before forecasting, the monthly sales series was decomposed into its trend, seasonal, and residual components and tested for statistical stability. Seasonality emerged as the dominant driver of sales, with a strength of **0.89** compared to **0.40** for the long-term trend. Sales typically peak in **December** (about **\$35,000** above average) and reach their lowest point in **February** (about **\$27,000** below average), indicating that the time of year has a much greater impact than the gradual upward trend.

The trend remained largely flat through 2015–2016 before rising steadily from **\$37,900** in mid-2016 to around **\$60,700** by late 2018, a **51%** increase. An **Augmented Dickey-Fuller (ADF) test** confirmed the series

was already stationary ($p = 0.0003$), so additional differencing was unnecessary and would have reduced forecast quality rather than improved it.

8. Forecasting Results

Three fundamentally different forecasting approaches were built and compared: SARIMA, a classical statistical model; Prophet, an industry-standard forecasting tool originally built by Facebook; and XGBoost, a machine-learning model adapted for time series through engineered features. To ensure a fair test, every model was evaluated on data it had never seen during training, using a strict chronological split rather than a random one — mixing past and future data during evaluation is a common and serious mistake in time-series work that this analysis specifically avoided.

A single three-month hold-out test is easily swayed by one unusual period, so a more rigorous check — a rolling-origin backtest that refits each model six separate times on expanding windows of history, pooling 18 forecast points in total — was used to make the final decision. The results:

Model	MAE	RMSE	MAPE
SARIMA	\$13,322	\$16,378	17.7%
Prophet	\$13,513	\$15,913	18.0%
XGBoost	\$14,724	\$17,424	20.4%

Rolling-origin backtest results, pooled across 6 origins (18 forecast points per model). Lower is better on every column.

SARIMA and Prophet perform almost identically on this more reliable test — SARIMA edges ahead on MAE and MAPE, Prophet is marginally better on RMSE, a difference well within normal noise. XGBoost trails both on every metric. Notably, a naive single-window test would have favored XGBoost (15.8% MAPE), which is exactly why the more rigorous backtest was used instead of a single lucky or unlucky three-month sample.

SARIMA was selected for production. At statistically equivalent accuracy to Prophet, it offers two properties a business forecast needs: its errors pass a formal test confirming they behave like random noise rather than a missed pattern (Ljung-Box $p = 0.44$), and it produces **calibrated confidence intervals** from a transparent statistical model — something XGBoost does not provide at all. Prophet remains a fully credible alternative and would be the easier of the two to extend later with holiday or promotional effects.

Three-month forecast (SARIMA, January–March 2019)

Month	Forecast	Low (95%)	High (95%)
January 2019	\$50,101	\$23,170	\$77,031
February 2019	\$35,740	\$8,812	\$62,669
March 2019	\$69,683	\$42,760	\$96,606

SARIMA point forecast with 95% confidence range, refit on the full four years of history.

In plain terms: sales are expected to dip in January and February, consistent with the seasonal pattern already established, before rebounding in March. The confidence ranges are wide relative to the point estimate — a natural consequence of forecasting from only four years of history — and should be read as a plausible range rather than a single guaranteed number.

9. Product & Region Forecasts

The production forecasting model was applied separately to Furniture, Technology, Office Supplies, West, and East, so that growth could be assessed at the level planning decisions actually happen. To isolate real year-over-year growth from the seasonal Q4-to-Q1 dip that affects every segment equally, each forecast was compared against the same calendar quarter one year earlier rather than the immediately preceding quarter.

Segment	Same Qtr, Last Year	Forecast, Next Qtr	YoY Growth	CI Width*
West	\$50,541	\$62,828	+24.3%	125%
Office Supplies	\$42,972	\$60,417	+40.6%	146%
Technology	\$55,690	\$66,551	+19.5%	187%
East	\$18,043	\$53,248	+195.1%	200%
Furniture	\$23,598	\$24,365	+3.3%	253%

**CI Width is the 95% confidence interval expressed as a percentage of the point forecast — a rough guide to how much to trust the growth figure. Sorted by growth rate.*

West shows the most credible growth story: +24.3% year over year, paired with the **tightest confidence interval of any segment** (about 125% of the point forecast, versus 146–253% for the rest) — the forecast a business could act on with the most confidence. **Office Supplies** is a strong second, at +40.6% growth with a moderately tight interval.

East shows the largest headline growth, +195.1%, but this figure needs context rather than face-value acceptance: its comparison quarter (Q1 2018) was unusually weak, and its confidence interval is roughly twice as wide as the forecast itself — among the least certain of the five. This is best treated as an upside scenario worth monitoring, not a number to commit inventory against directly.

Furniture is the weakest performer on both counts: only +3.3% growth, and simultaneously the widest confidence interval of any segment (253%) — a weak growth signal and the least certain one, arguing for holding stocking levels steady rather than expanding.

10. Anomaly Detection Findings

Two independent methods were used to flag weeks where sales looked unusual against the expected pattern, across all 209 weeks in the dataset: a rolling Z-score test (flagging any week more than two standard deviations from its trailing 13-week average) and an Isolation Forest model (a machine-learning method that flags unusual combinations of sales level and week-over-week change). Together they identified 19 distinct weeks: 11 by Z-score, 11 by Isolation Forest, and only 3 flagged by both.

The three weeks both methods agree on are the highest-confidence anomalies in the dataset, and each has a plausible, evidence-backed explanation rather than a guessed one:

Week	Sales	Order Lines	Flagged By	Likely Driver
Jul 26, 2015	\$21,590	60	Both methods	Broad-based demand
Sep 13, 2015	\$29,959	97	Both methods	Broad-based demand
Mar 22, 2015	\$37,704	54	Isolation Forest only	Large transactions
Dec 2, 2018	\$35,999	126	Both methods	Broad-based demand

Top anomalies by evidence strength. “Order Lines” measures how many individual line items made up the week’s sales — a high count indicates broad customer demand rather than a few large transactions.

July 26, 2015 and September 13, 2015 both show high order-line counts (60 and 97, well above the typical week), consistent with genuine demand surges around summer clearance and the September back-to-school and Labor Day period. December 2, 2018 similarly shows high order-line counts (126) consistent with the Cyber Monday and holiday shopping peak. All three line up with recognizable points on the retail calendar, which is exactly the kind of independent corroboration that should increase confidence in a flagged anomaly.

The single largest sales week in the entire four-year dataset — March 22, 2015, at \$37,704 — was flagged only by Isolation Forest. It falls too early in the data for the Z-score method to evaluate at all, since that method needs 13 weeks of history before it can score anything. Its evidence (54 order lines against a \$698 average order value, well above the \$206 typical average) suggests this week was driven by a handful of unusually large transactions rather than broad customer demand — a genuinely different kind of anomaly from the three above.

The two methods have real, complementary blind spots rather than one simply being “better.” Z-score cannot evaluate the first 13 weeks of any time series and is sensitive to the choice of that window length. Isolation Forest can overreact to sharp percentage swings when sales levels start out very low, as happened with several early-2015 weeks during the business’s ramp-up period. A monitoring system that uses both, and treats agreement between them as a confidence signal, is more robust than relying on either alone.

11. Product Segmentation Results

The 17 product sub-categories (such as Chairs, Phones, Copiers, and Paper) were grouped into four demand segments using K-Means clustering, based on four features: total sales volume, year-over-year growth, month-to-month volatility (relative to each product's own scale, not raw dollars), and average order value. The number of segments (four) was chosen using two independent statistical checks — the Elbow Method and the Silhouette Score — rather than picked arbitrarily.

Segment	# Products	Avg Sales	Avg Growth	Volatility	Avg Order Value
High Volume, Stable Demand	8	\$206,926	+21.5%	0.76	\$362
Low Volume, Stable Demand	6	\$37,371	+16.0%	0.72	\$50
High-Value, Growing Demand	2	\$167,743	+46.3%	1.50	\$1,931
Volatile, High-Growth Outlier	1	\$46,420	+192.8%	2.03	\$252

Cluster profiles. Volatility is a scale-free measure (0 = perfectly steady month to month); higher means more erratic.

High Volume, Stable Demand (Accessories, Appliances, Binders, Bookcases, Chairs, Phones, Storage, Tables) is the revenue backbone of the business, accounting for **73.2% of total four-year revenue** with moderate growth and the lowest volatility of any cluster — large, dependable, and well suited to automated replenishment.

Low Volume, Stable Demand (Art, Envelopes, Fasteners, Furnishings, Labels, Paper) covers smaller, lower-priced product lines with the steadiest demand pattern of the four groups — not a growth priority, but not a risk either.

High-Value, Growing Demand (Copiers, Machines) is defined by infrequent but large purchases — an average order value of \$1,931, more than five times any other cluster — combined with the strongest sustained growth rate (+46.3%) outside the single volatile outlier below.

Volatile, High-Growth Outlier (Supplies, standing alone) shows the highest average growth of any product group (+192.8%), but this figure is misleading at face value: it traces to a single collapse-and-rebound — sales fell 87% in 2016 before rebounding 653% in 2017 — rather than steady expansion, and this product line also carries the highest volatility in the portfolio.

12. Business Recommendations

Inventory planning

Set stocking policy by demand segment rather than uniformly. Use automated, moderate-safety-stock replenishment for the High Volume, Stable Demand cluster, which alone represents nearly three-quarters of total revenue and shows the lowest volatility in the portfolio. Use lean, low-safety-stock policies for the Low Volume, Stable Demand cluster, where demand rarely deviates from its own average. For the High-Value, Growing Demand cluster (Copiers, Machines), avoid holding large physical inventory of expensive, infrequently-sold items — a low-stock or make-to-order arrangement with reliable supplier lead times fits this demand pattern better. Supplies, as a volatile single-product outlier, warrants manual, short-cycle review rather than a standing policy of any kind.

Forecasting strategy

Adopt SARIMA as the production forecasting model, refit monthly as new sales data arrives, with Prophet documented and available as a fallback given its statistically comparable accuracy. Treat West and Office Supplies as reasonably confident candidates for proactive stocking increases given their combination of solid growth and tight confidence intervals; treat East's forecast as an upside scenario to monitor rather than a number to commit against, given its wide interval.

Anomaly monitoring

Implement a tiered alert system built on both detection methods together. Weeks flagged by both the Z-score and Isolation Forest methods should trigger immediate review, since the evidence in Section 10 shows these are consistently genuine, broad-based demand events. Weeks flagged by only one method warrant a lighter-touch check rather than an automatic escalation — this balances catching real shifts in demand against the alert fatigue that comes from over-flagging.

Demand segmentation

Use the four clusters identified in Section 11 as the basis for setting differentiated reorder rules, freeing up working capital currently tied to low-volatility, low-growth product lines while protecting availability for the faster-growing, higher-value segment. Revisit segment membership periodically as new sales data accumulates, since a cluster analysis run once on a fixed historical window is a snapshot, not a permanent classification.

Operational improvements

Build supply chain and staffing plans explicitly around the September back-to-school window and the November–December holiday peak, which this analysis confirms as the most reliable and largest seasonal events in the business, repeating in every year studied — rather than treating each year's peak as a surprise to react to after the fact.

13. Business Impact

This project is a historical analysis of four years of past sales, not a live deployment with measured outcomes, so the impact described here is directional and grounded in the patterns found in the data — not a projected dollar figure, which would not be honest without a live test.

- A calibrated forecast with a known confidence range gives inventory planning a defensible basis for the two costly failure modes of retail stocking — tying up capital in unsold inventory, and losing sales to stockouts — particularly ahead of the November–December peak, which this analysis shows is worth more than several other months combined.
- Segment-specific stocking policy means capital is not spread evenly across products with very different demand behavior; this creates room to reduce buffer stock on the stable, low-volatility majority of the catalog while protecting the smaller, faster-growing, higher-value segment.
- A two-method, tiered anomaly-alert design reduces the risk of both missing a genuine demand shift and burying staff in false-positive alerts from a single, overly sensitive method.

These effects follow logically from the patterns identified in this analysis; the natural next step, described in Section 15, is to validate them against real outcomes once the forecasting model and dashboard are put into active use.

14. Project Limitations

This analysis is only as strong as its assumptions, and several are worth stating plainly rather than glossing over.

- Four years of history is a real constraint. It is enough to establish an annual seasonal pattern, but not enough to be highly confident about slow-moving, multi-year trends, and some product sub-categories — Copiers, for instance, with only 66 orders across four years — have limited data to draw conclusions from.
- The forecasting models assume the near future broadly resembles the recent past. None of them can anticipate a structural change: a new competitor, a pricing shift, a supply chain disruption, or a broader economic downturn would not be reflected in a forecast built purely from historical sales.
- External factors that plausibly influence retail demand — promotions, marketing campaigns, competitor pricing, weather, and macroeconomic conditions — are not part of this dataset and are not modeled. The seasonal pattern captured here reflects the calendar, not these external drivers.
- Product segmentation was performed once on a fixed four-year window. Cluster membership, particularly the boundary between “growing” and “stable,” could shift meaningfully as more recent data becomes available.
- The Supplies cluster's extreme growth figure rests on a single product line with no other cluster members to compare against — conclusions there are inherently less robust than for the other three segments.

- Anomaly detection thresholds (the 13-week Z-score window, the 5% contamination rate used by Isolation Forest) were chosen using evidence from this specific dataset and time period; they would need to be re-validated before being applied to a different business or a different span of time.

15. Future Improvements

- Extend the historical window as more sales data accumulates — additional years would sharpen trend estimates and give the seasonal models more annual cycles to learn from.
- Incorporate external regressors — planned promotions, marketing spend, pricing changes, additional holidays — into the forecasting model. Prophet in particular is well suited to this kind of extension without a full model redesign.
- Automate model retraining on a monthly cadence as new sales data arrives, paired with drift monitoring that flags when forecast accuracy has degraded enough to warrant a closer look.
- Move the dashboard from local execution to a hosted deployment with scheduled data refreshes, so business users can check current forecasts without running the project locally.
- Extend anomaly detection toward real-time or near-real-time monitoring, so unusual weeks are surfaced as they happen rather than discovered in a retrospective analysis.
- Revisit product segmentation on a regular cadence — quarterly, for instance — rather than treating the four clusters identified here as a permanent classification.

16. Conclusion

Four years of Superstore sales data yield a consistent, well-supported picture: a business with strong, repeatable seasonality, a growth trend concentrated in its most recent two years, one region and one category carrying a disproportionate share of revenue, a handful of genuinely unusual sales weeks worth a closer look, and a product catalog that behaves less like one business and more like four distinct ones once demand patterns are actually examined.

Every finding in this report, and every number quoted, traces back to a specific calculation in the accompanying analysis notebook and can be reproduced from the underlying data. The forecasting model, anomaly screen, and product segments described here are built to be used, not filed away — the interactive dashboard accompanying this report puts all of them in front of the people who need to act on them.